

Homes, Heat and Healthy Kids Lothian Cohort (H3K Lothian)

Statistical Analysis Plan

Tim Wilding ¹, Caroline Fyfe ², Hannah Law ¹, Eleanor Harrison ^{1,3}, Vittal Katikireddi ⁴, Olivia Swann ^{1,5}

¹ Centre for Medical Informatics, Usher Institute, University of Edinburgh

² School of Geosciences University of Edinburgh

³ School of Informatics, University of Edinburgh

⁴ School of Health & Wellbeing, University of Glasgow

⁵ Department of Child Life and Health, University of Edinburgh

Abstract

Background

Cold, damp homes are associated with acute respiratory infections (ARIs) in preschool children. New data infrastructure in Scotland now permits linkage between healthcare data and individual properties, creating opportunities to explore this association in more detail.

Objective

The Homes, Heat, and Healthy Kids Lothians Cohort (H3K Lothian), will be established as an exploratory and developmental dataset within the wider Homes, Heat and Healthy Kids (H3K) programme. The aims are to develop and refine data linkage and cohort construction, finalise outcome definitions, develop infrastructure for multilevel count models and latent state models and examine associations between property energy efficiency and burden of acute respiratory infections.

Methods

We will construct a de-identified birth cohort of all children under 5 years old born in the NHS Lothian region from 01/01/2012- 31/12/24. We will link birth records with property level energy efficiency data, hospital admissions, GP attendances, prescriptions, prepayment smart meter data, environmental and climate data. The cohort database will be held in, and accessed via, the DataLoch Trusted Research Environment.

Descriptive statistics will be produced for the cohort over time. Multi-level models will explore the association between ARIs counts and home energy efficiency. Finally, we will develop a hidden Markov model to explore changes in the state of respiratory children's respiratory health with reference to the energy efficiency of their home.

Conclusion

The Homes, Heat, and Healthy Kids Cohort H3K Lothian will link administrative health data to longitudinal property level and environment exposures for the first time, and lay the groundwork for causal epidemiological analysis in a national dataset.

Contents

Abstract	1
Background to problem	3
Background to the Homes, Heat and Healthy Kids study	4
Objectives.....	4
Role of the H3K Lothian Cohort within the wider H3K Programme.....	4
Database Design	5
Population and follow up.....	5
Datasets and Variables	6
Missing data	14
Data Linkage	14
Database Management and Documentation.....	17
Statistical Analysis.....	17
Exposure definition.....	17
Outcome Definition	18
Covariates.....	19
Descriptive Analyses	19
Household Clustering of Respiratory Infections.....	20
Multilevel Regression Models.....	20
Hidden Markov Model Development	21
Limitations of H3K Lothian dataset.....	24
Patient and public involvement and engagement (PPIE).....	24
Dissemination.....	24
Conclusion	25
Statement on conflicts of interest	25
Funding.....	25
Ethics statement.....	25
Data availability statement	25
References.....	26

Background to problem

Acute Respiratory Infections (ARIs) are the leading cause for hospital admissions ^{1,2} in preschool aged children across the United Kingdom and are associated with cold, damp housing ³. ARIs are caused by viral or bacterial infection of the upper or lower respiratory tract ⁴⁻⁶ and manifest as respiratory issues including cough, wheeze or shortness of breath ⁷. Children's immature immune systems and smaller airways mean inflammation can block the respiratory tract more easily, with subsequent tissue damage and impaired lung function and lung development ^{8,9}. Childhood respiratory infections can have lifelong consequences, with increased risk of asthma and premature death ^{10,11}.

Preschool children spend much of their time at home. If the home environment is poor, this will increase exposure to environmental triggers such as low temperatures (below 18C), indoor air pollutants, damp and mould ^{12,13}. The World Health Organization's (WHO's) Housing and Health Guidelines highlights the importance of structural integrity and energy efficiency of houses in determining indoor thermal comfort, air quality, and the impacts on an individual's long term health¹⁴.

Socioeconomic pressures also play a critical role in a household's ability to maintain thermal comfort and when substantial enough, can result in fuel poverty. In Scotland, fuel poverty is defined as a household requiring to spend more than 10% of their adjusted income, after housing costs, on fuel to maintain a satisfactory heating regime and for their remaining adjusted income to be insufficient to maintain an acceptable standard of living ^{15,16}. When this fuel expenditure reaches 20% of a household's adjusted income, a household is deemed to be experiencing extreme fuel poverty. According to the Scottish Housing Condition Survey 2023, 34% of Scottish households were living in fuel poverty, and 19.4% extreme fuel poverty ¹⁷. Although income poverty is a key contributing factor to fuel poverty, in 2024, 22% of fuel poor Scottish homes were not income poor. Living in an energy efficient home does not necessarily protect from fuel poverty; 32% of dwellings with an Energy Performance Certificate (EPC) rating of band C or above, and 48% of dwellings with an EPC rating of band D or below, were living in fuel poverty ¹⁷. The relationship between fuel poverty and the energy efficiency of a house is further influenced by fuel and housing policy, a volatile and fluctuating energy market, and the rising cost of living. Consequently, a household can move in and out of fuel poverty as income and energy costs fluctuate over time.

Improving home energy efficiency (HEE) is one fundamental strategy to addressing underheating in homes and meeting Net Zero targets. The Scottish Government are therefore investing £1.8 billion to retrofitting existing homes to improve EPC ratings, with the goal of all homes rated band C or higher by 2033 ¹⁸. HEE interventions work by either reducing energy costs without improving the airtightness of a home (e.g. boiler replacement, solar panels) or increasing airtightness and improving heat retention (e.g. insulation or double glazing). Despite increasing the warmth of a home, interventions that improve airtightness may increase ARI risk due to trapping indoor air pollutants ¹⁹ and decreasing ventilation resulting in increased indoor humidity, damp and potential for mould growth ²⁰. Although there is a growing body of evidence demonstrating that HEE interventions that increase airtightness worsen respiratory health ²¹⁻²⁶, the National Institute for Health and Care Excellence (NICE) reported there is still a *"lack of rigorous, UK based evidence on the degree to which different [HEE] interventions modify the risk of cold temperature-related illnesses"* ²⁷. Although HEE of Scottish homes has increased from 52% in 2021 to 56% in 2024, fuel poverty has also markedly increased, by 9.1%, in the same time period ^{17,28}. As a result, defining and identifying underheated homes is difficult. Previous studies were limited by difficulties in sufficiently linking healthcare and housing data ²⁹. Consequently, quantification of the impact of underheated homes on the association between deprivation and preschool ARIs has not previously been possible.

Background to the Homes, Heat and Healthy Kids study

The Homes, Heat and Healthy Kids (H3K) study was established in 2024 to take advantage of recent developments in Scotland's national data infrastructure that allow healthcare records to be linked to individual residential properties using Unique Property Reference Numbers (UPRNs)³⁰. These advances created a new opportunity to examine how housing conditions influence child health at population scale. H3K brings together researchers across paediatrics, epidemiology, housing, energy and data science to link routinely collected healthcare data with housing, energy efficiency, environmental and socioeconomic datasets. The programme was designed as a phased research platform: an initial cohort using regional data (H3K Lothian) to develop linkage methods and analytical approaches, followed by a national cohort (H3K Scotland) assembled within the Public Health Scotland National Safe Haven to address the study's core policy and causal research questions (Figure 1). The H3K Scotland dataset will contain the H3K Lothian dataset as a subset.

Objectives

The H3K Lothian cohort is an exploratory and developmental dataset within the wider Homes, Heat and Healthy Kids (H3K) programme. It provides early access to linked healthcare and housing data while the national cohort is being assembled within the Public Health Scotland National Safe Haven.

The objectives of the H3K Lothian cohort are to:

1. Develop and refine data linkage and cohort construction methods.
2. Finalise ARI definitions across healthcare settings for 0-5 year olds.
3. Develop the analytical approach for multilevel count models and latent state models.
4. Examine associations between home energy efficiency and ARI burden in 0-5 year olds.

Role of the H3K Lothian Cohort within the wider H3K Programme

The H3K Lothian dataset is exploratory and does not include the property-level energy consumption or home energy efficiency intervention data required to define household energy deficit or to estimate policy counterfactual scenarios. Accordingly, analyses undertaken in DataLoch are exploratory in nature, and are not intended to provide definitive estimates of the attributable risk of underheating. Definitive exposure modelling (energy deficit), mediation analysis, and natural experiment evaluation will be conducted in the forthcoming national cohort (H3K Scotland) within the Public Health Scotland National Safe Haven, with a separate statistical analysis plan.

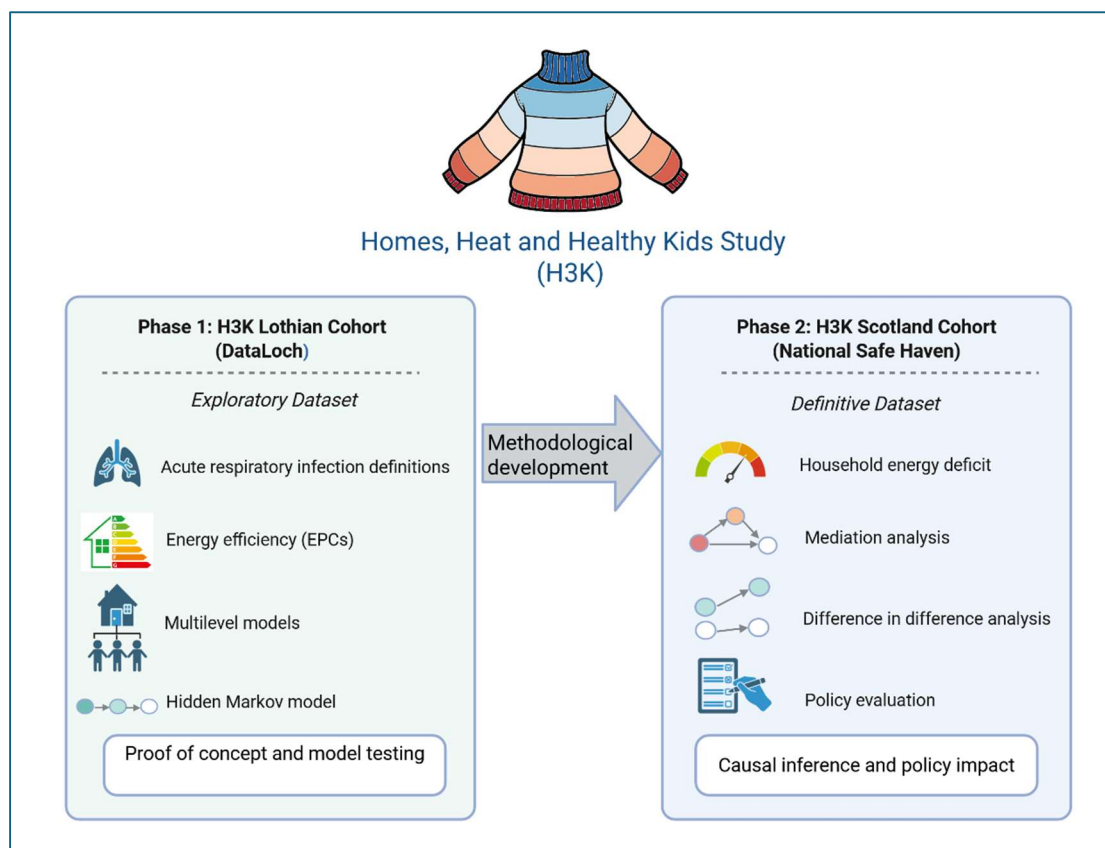


Figure 1. Home, Heat and Healthy Kids Study cohorts

Database Design

Population and follow up

H3K Lothian is a retrospective birth cohort of all children born in NHS Lothian from 01/01/12 – 31/12/19 who are registered with a GP, followed until their fifth birthday, migration from the region, death, or the end of the study period (31/12/24). Healthcare records (primary care, secondary care and community prescriptions) will be linked to homes using the Community Health Index (CHI):UPRN file³⁰ allowing children to be located within properties and neighbourhoods. The cohort structure will allow modelling of acute respiratory infections at three hierarchical levels: level 1: child, level 2: property, level 3: neighbourhood (census output area). The cohort database will be held in, and accessed via, the DataLoch Trusted Research Environment ³¹.

Inclusion criteria:

- Born between 1st January 2012 and 31st December 2018
- Valid pseudonymised patient identifier enabling linkage across datasets
- Continuously resident within the NHS Lothian area from birth until 5th birthday or censorship from the study, with no recorded out-of-region addresses or address gaps. Moves within Lothian are permitted. A four-month grace period after birth without an address will allow for time to register the birth.
- Linked to at least one residential property through UPRN

Exclusion criteria:

- Born outside of the study period
- No valid UPRN-linked address during follow-up

- Evidence of being born outside NHS Lothian
- Implausible or conflicting residential timelines that cannot be reconciled

Followed up from birth until the earliest of:

- Fifth birthday
- Migration out of NHS Lothian
- Death
- End of study period

Datasets and Variables

H3K Lothian links several health, environmental and geospatial datasets outlined in Table 1.

SMR01 - General Acute Inpatient and Day Case Admissions

The Scottish Morbidity Record 01 (SMR01 ³²) is a national administrative dataset of episode-level data on general acute inpatient and day case admissions across non-maternity, non-psychiatric hospital care. The SMR01 record contains demographic and administrative information such as the patient's Community Healthy Index (CHI) number, age, sex, admission and discharge dates. The record contains up to six diagnostic fields; a main diagnosis field and up to five other condition fields. These condition fields are entered using the International Classification of Diseases, 10th Revision (ICD-10) code. The clinical coding is undertaken by specialist coders who translate the written diagnoses to the appropriate ICD-10 codes after discharge. ICD-10 coding accuracy is limited by clinician's documentation and can vary between hospitals and coders.

SMR02 -Scottish Birth Record

Scottish Morbidity Record 02 (SMR02 ³³) is the main administrative dataset recording clinical and demographic data for maternity inpatient and day case events. Records for SMR02 are submitted by NHS hospitals whenever a pregnancy ends in delivery or discharge. SMR02 contains a wide range of data on the mother and the birth. Fields for the mother include age, height, smoking history and obstetric history. Data about the birth includes gestational age, birth weight, delivery methods and birth outcomes (e.g. stillbirths). SMR02 is episode-based, meaning that multiple records can exist for the same mother across antenatal and delivery care. Gestational age is the completed number of weeks of gestation at birth and is based on the estimated date of delivery from antenatal records. The estimated gestation is used to determine foetal maturity at birth and can be compared to birth weight to determine where the birth weight is relative to gestational age.

Table 1. Key data sources uses in this study

Dataset	Source	Description	Key Variables	Dates available	Granularity of data
Demographics	DataLoch Core Demographics	Demographics from GP registration data	Date of birth, ethnicity	01/01/12 – 31/12/24	Child
Scottish Morbidity Record (SMR) 01	Public Health Scotland	SMR01 includes all general day case and inpatient admissions in Scotland. Admissions are filtered to the admissions that have relevant ICD10 codes	ICD10 condition codes, admission dates	01/01/12 – 31/12/24	Child
Scottish Morbidity Record (SMR) 02	Public Health Scotland	SMR02 includes all day case and inpatient admissions to the maternity specialities including births.	Gestation, birth weight, mode of delivery, maternal smoking history	01/01/12 – 31/12/24	Child
GP Record Data	Albasoft	Includes all GP admissions in the NHS Lothian region	Date of treatment, Read v2 codes	01/01/12 – 31/12/24	Child
Community Prescription Data	Prescribing Information Systems	All prescriptions of Clarithromycin, Amoxycillin and Salbutamol in the NHS Lothian region	Prescription name, date	01/01/12 – 31/12/24	Child
Death Record	National Records of Scotland	Nationally curated data for the date and cause of death	Cause of death	01/01/12 – 31/12/24	Child
Scottish Index of Multiple Deprivation	National Records of Scotland	Area deprivation by quintiles across income, employment, education, health, access to services, crime and housing. (2020 version)	SIMD for each address in Scotland	01/01/12 – 31/12/24	Postcode
Energy Performance Certificate (EPC)	Energy Saving Trust	Energy efficiency of the home when assessed	EPC rating, energy efficiency features, tenure	01/01/12 – 31/12/24	Property
uZero Fuel Poverty Risk Score	UrbanTide	Calculated using aggregated prepayment smart meter data, proportion of population on universal credit and individual property energy efficiency	Fuel poverty risk score (0-100), percentage of population on universal credit, modelled EPCs	01/06/24 – 01/06/25	Output area, postcode and property
Climate data	Centre for Environmental Data Analysis	Monthly climate data for nine meteorological indicators	Temperature, humidity	01/01/12 – 31/12/24	1 x 1 km grid
Outdoor air pollution data	Department for Environment, Food & Rural Affairs	Annual outdoor air pollution data for the concentrations of seven key pollutants	PM 2.5, PM10, NO2 etc.	01/01/12 – 31/12/24	1 x 1 km grid

Energy Performance Certificates (EPCs)

EPC data³⁴ provides a property-level view of energy efficiency assessments while protecting identity through a pseudonymised Unique Property Reference Number (UPRN). Each record represents an EPC inspection event and includes the pseudonymised UPRN (a stable token that supports longitudinal linking within the dataset), the inspection date (ideally in ISO 8601 format), a set of energy consumption indicators (for example, estimated annual kWh, electricity and/or gas consumption where available), and the Energy Performance Certificate information. EPC content typically includes the rating band (A–G) and the underlying numeric score (Standard Assessment Procedure [SAP], 0–100), with optional certificate identifiers and links to the source certificate if present. Although a property may be inspected multiple times, H3K Lothian uses a dataset consisting of the latest property inspections for each property and the EPC information is assumed to be constant over the study period.

The EPC dataset is designed to support analyses such as tracking changes in energy performance over time, benchmarking consumption against EPC ratings, and identifying cohorts for retrofit or targeted interventions. EPC consumption figures are often modelled estimates rather than metered use. Inspection dates can cluster around policy or market events, and some fields may be missing or updated across re-inspections.

EPCs are imperfect records of a dwelling's current state. The EPC ratings are based on the SAP/RdSAP model³⁵ and reflect the assessor's input choices. The EPC energy usage is an estimate of theoretical energy usage rather than actual energy usage. EPCs are produced when the dwelling is sold, let or built. They remain valid for up to 10 years and reflect transactional activity rather than a regular reviews. Many households will not have a current EPC and coverage will vary across areas and tenures, e.g., long-term owner-occupied homes are less likely to have a current EPC than a property that is let more frequently. Using the latest EPC certification means that results can be biased by certificate age (e.g. unobserved retrofits). EPC certification will not be missing at random because of the differential likelihood of a property receiving depending on tenure.

uZero Fuel Poverty Risk Score

The uZero Fuel Poverty Risk Score³⁶ is produced by UrbanTide and incorporates the pattern of electricity and gas meter top ups and disconnections to assign individual smart meters to different behaviour clusters (fuel poor or not fuel poor). The score ranges from 0–100 and was designed to predict the potential benefit of home energy efficiency measures in reducing fuel poverty in each output area (100 = most impact, 0 = least impact). The score is calculated as the min-max normalised mean value of the following features after they are standardised:

- Number of people on Universal Credit divided by the number of domestic UPRNs
- Proportion of fuel poor gas smart meters in the local super output area*
- Proportion of fuel poor electricity smart meters in in a local super output area*
- The mean increase in EPC energy efficiency score in an output area required to either meet the households' potential energy efficiency or at least Band C

* If there are <15 devices in the LSA, this is smoothed with the local authority number. The prepayment smart meter data underlying the uZero Fuel Poverty Risk Score is provided by the Data Communications Company³⁷ (the licenced monopoly regulated by Ofgem with responsibility for operating the technology infrastructure and data flow of UK smart meters). This smart meter data is supplied to UrbanTide already aggregated at the output area level. The uZero Fuel Poverty Risk Score contains three levels of information: output area level, post area level and property-level.

uZero Output Area Features

The Output Area (OA) is the smallest geographical level for which census estimates are provided³⁸ and in Scotland typically consists of ~50 households. The uZero OA dataset contains a small-area view of energy vulnerability, energy efficiency, and meter-based segmentation. In addition to the Fuel Poverty Risk Score described above, it also contains vulnerability measures including deprivation indices and proportion of households receiving Universal Credit. The dataset contains average EPC rating for each OA, from the EPC Open Dataset³⁹. Department of Energy Security and Net Zero energy and off-gas statistics⁴⁰ are used to build off-gas grid and household fuel type counts across each OA. Universal Credit counts are sourced from DWP's Stat-Xplore⁴¹.

Postcode Area Features

The uZero postcode area features file lists domestic energy characteristics and energy consumption at the postcode-level and is designed to support spatial analysis and targeting of resources. The dataset combines modelled data from EPC Open Data³⁹ (using the SAP methodology³⁵) with annual energy consumption statistics from the National Energy Efficiency Dataset (NEED) 2023 release⁴². The dataset also includes contextual indicators such as number of dwellings off the mains gas grid and income level from Experian data. The EPC-derived components characterise building fabric and modelled energy usage such as total consumption (based on floor area, current vs potential energy efficiency and the gap to EPC band C). The NEED-based components summarise actual energy use recorded by electricity and gas meters in the area. EPC values reflect modelled energy under standard conditions, while NEED reflects actual consumption, behaviour and weather patterns in 2023 by postcode. The file is constructed by aggregating dwelling-level EPC data values to the postcode level. NEED data is aggregated from the domestic meters associated with the postcode.

Property-level Features

Dwelling-level data (including EPC rating, mains gas, tenure, property type, insulation and heating system) are available for each property. If a current EPC is available through EPC Open Data³⁹ (using the SAP methodology³⁵) this is used. If a current EPC is not available for the property, this data is imputed by UrbanTide using the following approach. Where no valid EPC is available, a value is imputed using a machine learning model which predicts a value based on the EPC ratings of similar properties nearby, as well as using other data available for the property. Where other data is missing, this defaults to average of ratings in the postcode.

A version of the output area fuel poverty score is recalculated, now using the EPC C-Gap specific to the particular property if an EPC is available. If a current EPC is not available, this defaults to the output area Fuel Poverty Risk Score.

Air Pollution Data

Air pollution data ⁴³ is provided by the Department for Environment, Food and Rural Affairs (DEFRA) at 1x1 km resolution. The data is stored using the easting/northing grid reference system, and the centroid of each 1x1 square is used as the calculation point when determining distance to the grid square from any other point. Data is provided on an annual basis for the set of seven pollutants shown in

Table 2.

Table 2. Annual pollution measures available in study

Variable	Unit	Description
Benzene	mg m ⁻³	Annual mean concentration
NO ₂	mg m ⁻³	Annual mean concentration
NO _x	mg m ⁻³	Annual mean concentration
Ozone	Days	Number of days where daily max running 8-hr mean concentrations exceeds 120 ug m ⁻³ averaged over 3 years (confirm)
PM10	mg m ⁻³	Annual mean concentration
PM2.5	mg m ⁻³	Annual mean concentration
SO ₂	mg m ⁻³	Annual mean concentration

Three pollutants will be considered in the final model:

- PM2.5 – fine combustion-derived particles that penetrate deep into the alveoli
- Ozone – secondary photochemical oxidant with low correlation to primary traffic pollutants such as NO₂
- NO₂ – traffic-related reactive gas that is a widely used measure of combustion emissions

The choice of pollutants is based on relevance to ARIs ⁴⁴ while minimising the redundancy and multicollinearity suggested by the correlation matrix. The correlation matrix in Figure 2 shows the pollutants clustered into three groups – NO₂ and the traffic-related pollutants, Ozone is uncorrelated, and PM2.5 and the particulate-related pollutants. NO₂ is highly correlated with other traffic-related pollutants such as NO_x and benzene. Ozone is formed via photochemical reactions and often peaks in warmer months and is often lower at busy urban roadside sites.

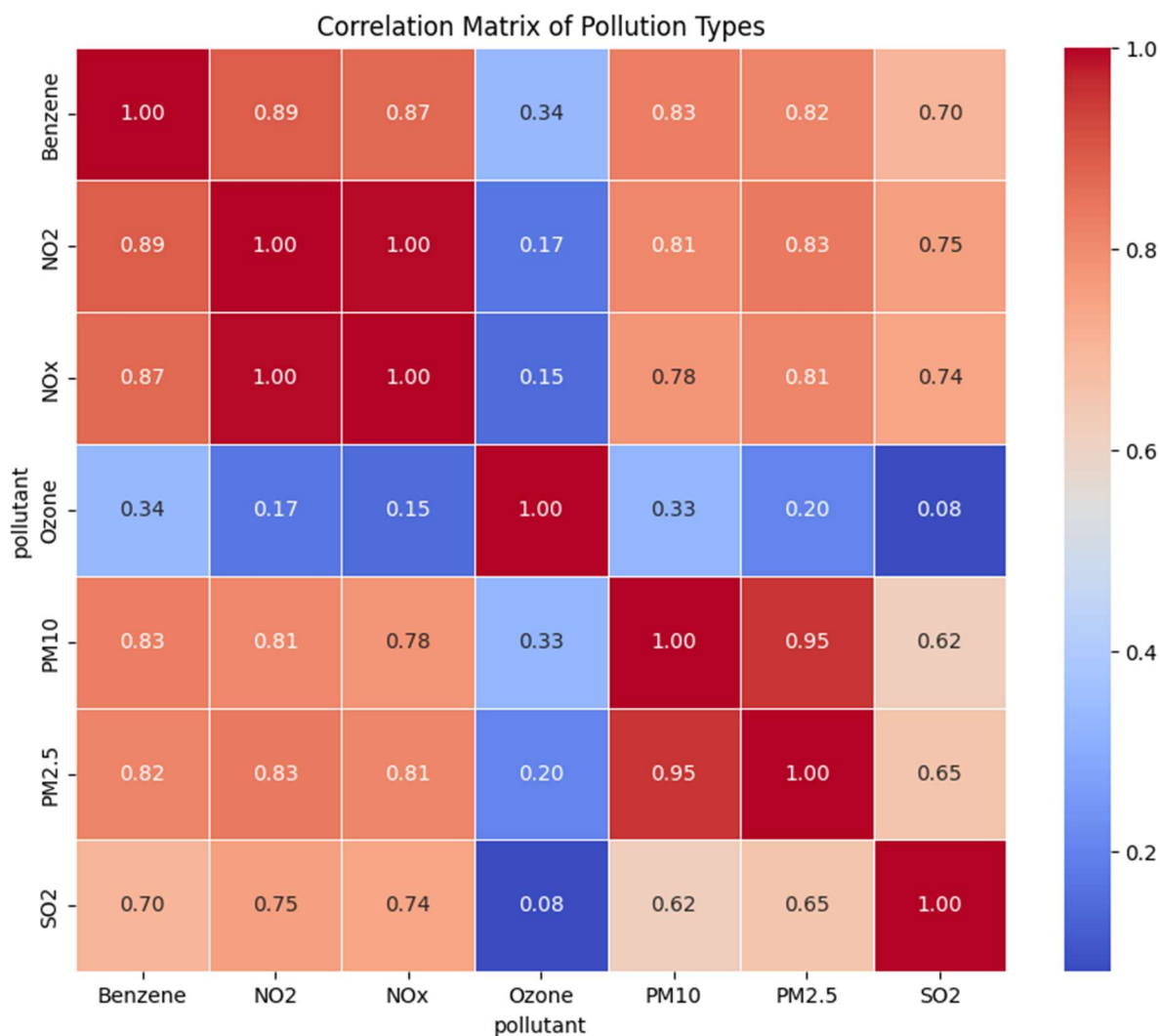


Figure 2. Correlation matrix of the pollutants across the different grid points in the UK from 2012-2024

PM2.5 penetrates the lungs more than the other particulate-based pollutants and can impair antiviral/antibacterial defences⁴⁵. Ozone is an oxidant that can increase inflammation of the airways and increase susceptibility to bacterial and viral respiratory infections⁴⁶. NO₂ can also cause inflammation of the airways particularly in young children⁴⁷.

Climate Data

Climate data⁴⁸ is provided by the Centre for Environmental Data Analysis (CEDA) at 1x1 km Resolution. The data is stored using the easting/northing grid reference system, and the centroid of each 1x1 square is used as the calculation point when determining distance to the grid square from any other point. Data are provided monthly for the set of nine climate indicators shown in .

Table 3.

Table 3. Monthly climate data available in study

Variable	Unit	Description
Ground frost	1.0	Number of days with ground frost when the minimum ground temperature fell below 0°C
Relative humidity	Percent	Near-surface relative humidity
Rainfall	mm	Total rainfall
Surface wind speed	ms ⁻¹	Average wind speed over month
Snow lying on ground	1.0	Number of days with snow lying
Hours of sun	Hours	Hours of sun per day
Temperature	°C	Average temperature over month
Minimum temperature	°C	Average daily minimum temperature
Maximum temperature	°C	Average daily maximum temperature

Three climate indicators will be considered in the final model:

1. Mean temperature
2. Ground frost
3. Relative humidity

The selection of indicators was informed by cluster analysis which revealed that mean temperature was highly collinear with the minimum and maximum temperature. A such, mean temperature was chosen as a single indicator. Ground frost days are an indicator of cold spells and should help capture heating requirements that may be missed in the average temperature measures. Snow lying on the ground covaries with the ground frost and is, therefore, a redundant measure. Relative humidity is the most direct proxy of the damp conditions that a household will face. Underheating when there is high relative humidity will foster mould and dampness because cold air has lower moisture capacity. Rainfall covaries with relative humidity and is a less direct indicator of damp.

Geospatial Reference

The Geospatial Reference file links children to different addresses throughout their life. Each record shows an identifier for a child with a starting and ending month for a particular home. The records are linked by pseudonymised fields including UPRN, climate grid, pollution grid, output area and postcode.

SIMD

SIMD estimates children's socio-economic status throughout their early life. Each record shows a pseudonymised identifier for a child with a starting and ending month for a particular SIMD decile ⁴⁹. SIMD ranks 6976 small Scottish areas across seven domains including income, employment, education, health, access to services, crime, and housing. Ranks for these seven domains are combined into an overall SIMD rank. Each data zone is ranked from most deprived (rank 1) to least deprived (rank 6976), split into deciles. The SIMD rank is an area-based measure of relative deprivation, which means that not every child in the area will themselves experience the same rate of deprivation. Rural areas are often larger with a more mixed picture of deprivation. To avoid circularity in health analyses, we will use the income and employment sub-indices of SIMD. This widely adopted approach isolates socioeconomic drivers without factoring in health.

Missing data

There are multiple sources of data linked together and that will give rise to multiple patterns of missingness. Demographics, Geospatial Reference, EPC, Environmental Data and Hospital admissions will all have distinct patterns of missingness:

- Demographics- Many children were not born in the Lothians region; birth records (e.g., birthweight, gestation) are absent for these individuals. Address records are incomplete or absent for a substantial subset, limiting geocoding and spatial linkage.
- Geospatial Reference- House-level records are incomplete; some addresses are not matched.
- EPC (Energy Performance Certificates)- Many geospatial records have no EPC; within-EPC fields (e.g., roof properties, secondary heating, and especially air tightness) have high within-record missingness.
- Climate/pollution- Near-complete with minimal gaps.

The following assumptions will be used about missingness and risk of bias. The data is expected to show deviations from Missing Completely at Random (MCAR, ⁵⁰) and frequent violations of Missing at Random (MAR), particularly:

- EPC availability is Missing Not at Random (MNAR): e.g., long-term owner-occupiers are less likely to have EPCs than renters, and tenure relates to housing conditions and socioeconomics.
- Address missingness may depend on deprivation or housing instability.

As such, complete-case analyses may be biased toward families with stable addresses, renters with EPCs, and locally born children.

Harmonising data and coding principles

“Unknown” values will be treated as a distinct category when it represents a specific state such as the tariff being “unknown”. Other variables will be marked as “NA” if they reflect lack of available data. Multiple Imputation (MI) will be used for variables with moderate levels of missing data where Missing At Random should be tenable (e.g. wall insulation), but variables with a high-level of missingness will be excluded from any final analysis (e.g. loft insulation).

Data Linkage

Creation of the H3K Lothian spine

The H3K Lothian spine will consist of all live-born children born in the NHS Lothian region within the study period with valid CHI and continuous residential UPRN for the first five years of their life. The spine will exclude properties with irreconcilable linkage. Children will be followed from birth up to the 5th birthday (or censoring/death/migration). Linkage from the H3K spine is shown in Figure 3.

Linking the H3K Lothian spine to health records

Scotland adopted the Community Health Index (CHI) in the 1970s ⁵¹. Every person registered with a GP in Scotland is allocated a 10-digit CHI number from a centrally maintained register. The register contains data on address, postcode, GP, date of birth, region of registration and, if relevant, date of death. This CHI number is the unique patient identifier used across all of NHS Scotland. The Scottish Morbidity Records (SMR) identify the patient with the CHI number. GP Visits and prescriptions also identify the patient by CHI number. Admissions to hospital are classified using ICD-10 codes with comorbidities also captured using these whereas GP attendances are classified using Read v2 codes.

Linking the H3K Lothian spine to environmental data

DataLoch stores the address history for patients in the Lothian region. On a monthly basis DataLoch get a series of files from Public Health Scotland CHI Linkage and Indexing team (PHS CHILI) used to match addresses to the correct UPRNs. Address entry in the various medical systems isn't standardised and is often entered in free text so a single property may have been entered onto the systems in multiple ways over the years due to typos, different naming conventions for flats etc. Any remaining unmatched addresses in DataLoch are run through the flap_lite tool ⁵² to check if that can identify UPRN's for any of those addresses. About 90% of the addresses in the DataLoch database are successfully matched to UPRNs.

Climate and air pollution data are available for 1x1km grids. These are linked to UPRN by determining the UPRN's easting/northing grid reference from the Ordnance Survey Address Base Premium ⁵³ dataset, find the closest grid square centroid in a straight line for climate and air pollution data and returning the data for that grid square.

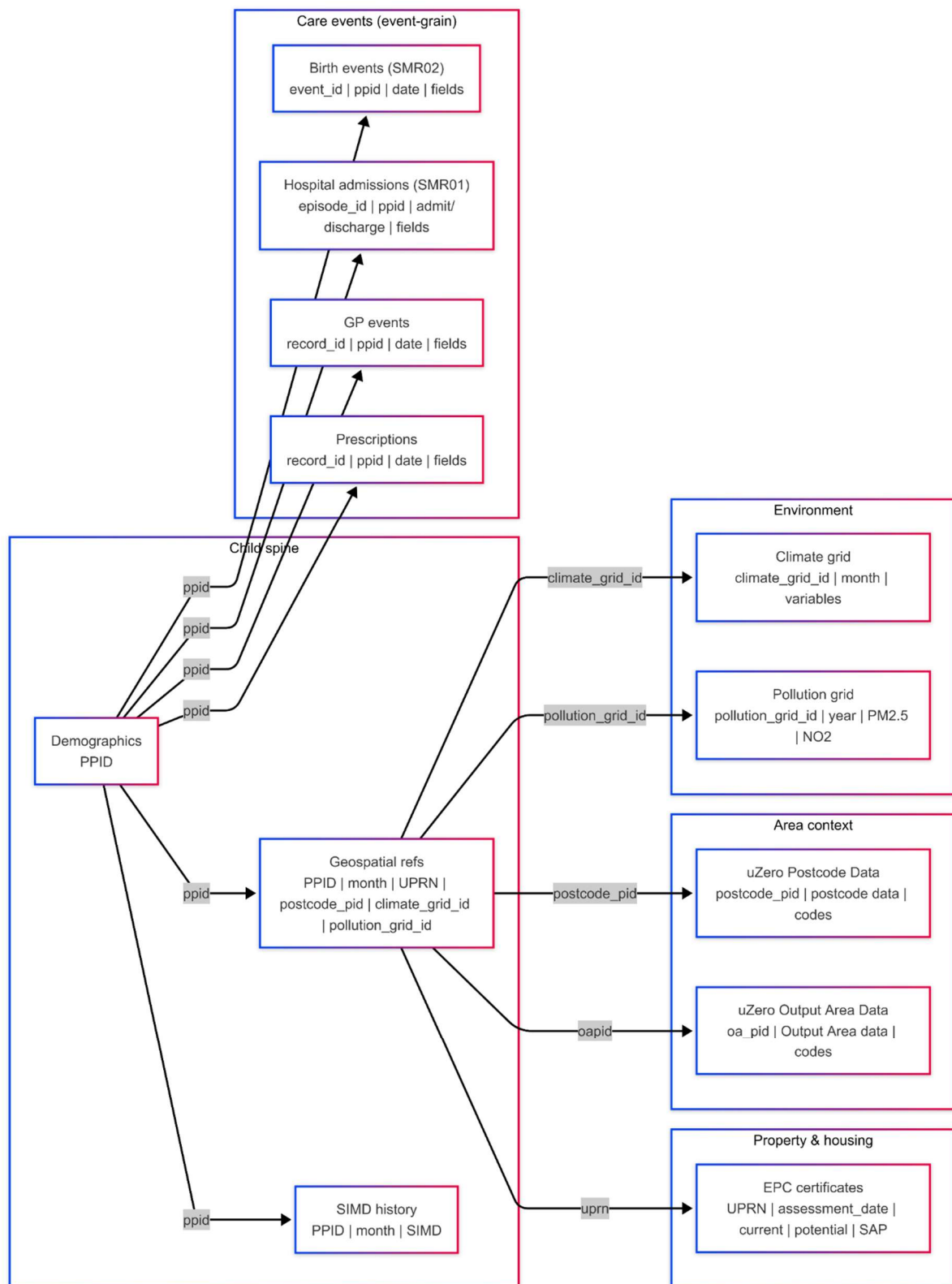


Figure 3. Diagram showing the linkages between the different datasets in H3K Lothian. All identifiers (ppid, uprn_pid, etc.) are pseudonymised. PPID = Pseudonymised Personal Identifier, UPRN = Unique Property Reference Number, PID = Pseudonymised Identifier, SAP = Standard Assessment Procedure, EPC = Energy Performance Certificate, SMR = Scottish Morbidity Records

Database Management and Documentation

A collection of tab-separated values (TSV) files is ingested into an in-process analytical database (DuckDB ⁵⁴) to create an analysis-ready population database. Ingestion proceeds into a “raw” schema that preserves source structure (field names, encodings, and missing-value conventions). DuckDB’s columnar, vectorised execution and efficient scanning of delimited files enable fast load-and-transform without requiring a separate database server. The data is transformed to correspond to tidy-data principles ⁵⁵ to facilitate statistical analyses. Transformations are orchestrated with dbt ⁵⁶ using the DuckDB adapter. The transformations adopt an ETL-style pattern use declarative SQL models to enclose the transform logic. The transform logic is compiled, executed, documented and tested as part of a version-controlled project. The data pipeline is organised into layered models:

- a staging layer standardizes sources,
- intermediate models apply deduplication, and conformance of demographic dimensions,
- a final layer of marts exposes clean, analysis-ready fact and dimension tables.

Custom tests enforce data quality, while documentation and lineage graphs provide transparent data provenance from raw TSVs to final analytical outputs.

Statistical Analysis

Exposure definition

The exposure of living in a cold home cannot be measured directly without access to temperature and humidity metrics for each property. As such, we considered different proxy indicators, as summarised in Table 4.

Table 4. Potential indicators of exposure to living in a cold house

Indicator	Strengths	Limitations
SAP score (0-100) from EPC	• Available at house level. • Indicates difficulty to heat the house and maintain warmth. • Indicates higher heating costs. • Standardised measure. • Absolute measure.	• Does not indicate heating affordability. • Measures energy efficiency not actual temperature. • EPCs have known measurement errors
uZero energy poverty risk score	• Available at house level. • A composite measure that includes energy efficiency and affordability. • Standardised measure.	• Derived from an area-based measure. • Relative measure.
Low EPC rating, low-income interaction.	• Available at house level. • Indicates efficiency and affordability of heating. • Absolute measure.	• Not a standardised measure – limiting external comparability. • May not be a sufficient number of houses eligible for inclusion. • Would need to develop, test and validate the measure before use.

Primary exposure: Home energy efficiency rating

After considering the quality and availability of data and the sensitivity of the indicators, it was decided that the primary exposure will be structural energy efficiency of the home, measured using the Standard Assessment Procedure (SAP) score (0-100) from the most recent Energy Performance Certificate (EPC) available for each property. Where available, modelled EPC values will be used to provide full population coverage. SAP score will be analysed as a continuous variable using flexible spline models. Secondary analyses will examine EPC bands and SAP quintiles.

H3K Lothian contains a single EPC for each home, which is the most recent one lodged. As such, EPC values will be treated as time-invariant structural characteristics of properties. Sensitivity analyses will explore restricting the cohort to children born after EPC inspection where inspection dates are available.

Secondary Exposure (Exploratory)

For recent years where available, the uZero fuel poverty risk score will be examined as an exploratory proxy for underheating of the home. This variable combines property EPC characteristics and prepayment smart-meter-derived patterns of top ups and disconnections at the local level. As this metric is only available for recent years, it will be used in restricted-period analyses only.

Outcome Definition

The outcome will be healthcare use for ARIs in children under 5 years old. Outcomes will be identified using harmonised code lists across healthcare settings:

- Hospital admissions for ARIs (ICD-10 codes)
- Primary care consultations for ARIs using Readv2 codes)
- Prescriptions for salbutamol, amoxicillin and clarithromycin (British National Formulary codes).

For descriptive analyses (see below), the primary outcome will be the number of ARI hospital admissions per child-month of follow-up, where a child-month represents one child contributing one month of observation time to the study. Secondary outcomes will be number of GP ARI attendances per child-month and prescriptions per child-month.

To improve interoperability, published code lists from the HDR UK phenotype library will be used to identify ARI admissions^{57,58} and GP attendances⁵⁹⁻⁶². Asthma is hard to diagnose in children under five years and cannot be reliably distinguished from an ARI. As such, we will also include ICD-10 and Read v2 codes for acute asthma. Repeated healthcare contacts within short time windows will be grouped into infection episodes to avoid double-counting, and incidences within a rolling window of seven days will be counted as a single incident. Sensitivity analyses will explore the impact of using five, seven or ten days as the cut off for a single episode.

Admissions will be considered as due to an ARI if an ICD-10 code from the ARI code list is present in any of the diagnosis positions 1-3 of the hospital admission record in SMR01. For an asthma admission, the ICD-10 code must only appear in diagnosis position 1. This will exclude diagnoses where asthma is only present as a comorbidity. GP encounters will be considered as due to an ARI if they include Read v2 codes from the ARI code list.

Admissions, GP attendances and prescriptions will also form the multivariate emissions of the Hidden Markov Model (see below). These signals will also be combined into a single costed healthcare episode to estimate time-varying costs of ARIs.

Covariates

A range of covariates representing child, household, housing, and neighbourhood characteristics will be included in descriptive summaries and regression models to account for variation in demographic, socioeconomic, and environmental context across the cohort. These covariates are selected based on data availability within the H3K Lothian dataset and relevance to early childhood respiratory health.

Child-level: Covariates will include sex, ethnicity (grouped into the six high-level Scottish Census categories), gestational age at birth, birthweight for gestational age, and age during follow-up. Age will be incorporated either as age-year bands or as a continuous time variable depending on the modelling approach. Seasonal and calendar time indicators (e.g. month or season of observation) will also be included to capture temporal patterns in respiratory infections. Our group have previously produced code lists for 22 chronic paediatric conditions, harmonised across primary and secondary care ⁶³. These will be used to identify groups of children with pre-existing comorbidities which may make them more susceptible to ARIs.

Property level: Covariates will include housing tenure (owner-occupied, private rented, social rented, or unknown), and property type (e.g. detached, semi-detached, terraced, or flat/apartment). Residential mobility during follow-up will also be captured using the number of changes of residence experienced by each child. Where siblings are identified in the dataset, when one sibling is documented as having moved, all others will presume to have moved on the same date.

Neighbourhood-level: Covariates will include area-level deprivation measured using the SIMD quintiles, local authority area of residence (City of Edinburgh, East Lothian, Midlothian, and West Lothian), energy poverty risk (uZero data), outdoor air pollution climate.

These variables will be used to describe cohort characteristics and to explore how respiratory infection burden varies across demographic, housing, and neighbourhood contexts in the Lothian region. Covariate availability and completeness will be assessed during descriptive analyses, and categories may be collapsed where necessary to ensure stable estimates.

Descriptive Analyses

Descriptive statistics will be used to characterise the cohort and assess how representative it is of the wider population of children in NHS Lothian. We will summarise demographic, socioeconomic, and housing characteristics of the cohort using contingency tables and summary measures. The primary outcome for descriptive analyses will be the number of acute respiratory infection (ARI) hospital admissions in early childhood, summarised across age-year bands (≤ 1 , $>1 \leq 2$, $>2 \leq 3$, $>3 \leq 4$, and $>4 \leq 5$ years) and across the overall 0–5 year period. Within each age band, children will be categorised according to the number of ARI admissions experienced (0, 1, 2, or ≥ 3 infections). For children who are censored before their fifth birthday, results will be reported up to the last fully observed age band and partial follow-up will be noted.

For each age band and for the overall 0–5 year period, the proportion of children in each infection-count category will be reported across the following characteristics:

Child characteristics

- Sex (as recorded in healthcare records)
- Ethnicity (grouped into the six high-level Scottish Census categories)

Socioeconomic context

- SIMD quintile (based on the child's postcode at the start of each age-year band)
- Local authority (City of Edinburgh, East Lothian, Midlothian, West Lothian)

Housing characteristics

- Housing tenure (owner-occupied, private rented, social rented, unknown)
- Property type (detached, semi-detached, terraced, flat) derived from EPC data
- Energy efficiency:
 - SAP score (continuous)
 - EPC band grouped as C or above vs D or below

Housing precarity

- Number of residential moves during follow-up (0, 1, ≥ 2 moves)

Neighbourhood energy context

- Risk of fuel poverty based on uZero output areas data, grouped into quintiles.

These descriptive analyses are intended to characterise the distribution of respiratory infection burden across early childhood and to explore how this burden varies across demographic, socioeconomic, housing, and neighbourhood contexts in the Lothian region. Outputs are purely descriptive and will be used to inform the specification of subsequent modelling analyses.

Household Clustering of Respiratory Infections

To explore potential household clustering of respiratory infections, we will examine the distribution of ARI events among children residing in the same property. Using the pseudonymised UPRN, we will identify properties in which more than one child from the cohort resides during overlapping time periods. Descriptive analyses will summarise the number of children per household, the proportion of households in which more than one child experiences at least one ARI admission during the study period, the distribution of ARI counts among children sharing the same household. These analyses will provide an initial assessment of within-household clustering of respiratory infections and will inform the specification of subsequent multilevel models that account for clustering at the property level.

Multilevel Regression Models

Associations between energy efficiency (SAP) and ARI burden will be examined using multilevel count regression models. These models allow estimation of infection rates while accounting for the hierarchical structure of the data and repeated observations over time.

The primary outcome will be the number of ARI hospital admissions occurring within each child-month of follow-up. As ARI admissions are recurrent events, modelling the count of events over time allows efficient use of the full follow-up period while accounting for differences in duration of observation between children. A multilevel Poisson regression model will initially be used to estimate incidence rate ratios (IRRs) for ARI admissions associated with energy efficiency. The model will include random effects to account for clustering at multiple levels (repeated observations within child, children nested within homes, properties nested within neighbourhoods). This hierarchical structure reflects the fact that respiratory infection risk may be correlated among children living in the same household and among households located in the same neighbourhood. If evidence of overdispersion is observed (i.e. variance exceeding the mean of the count outcome), alternative specifications such as multilevel negative binomial models will be explored.

The exposure variable will be the SAP score, modelled as a continuous variable using restricted flexible splines to allow for potential non-linear relationships between energy efficiency and infection risk. Secondary analyses will examine EPC band grouped as C or above versus D or below. Models will incorporate covariates at multiple levels, (outlined above in Covariates section), reflecting child, household and neighbourhood characteristics. Results will be reported as incidence rate ratios (IRRs) with corresponding confidence intervals, representing the relative change in ARI admission rates

associated with differences in residential energy efficiency. To explore the non-random pattern of missingness in EPCs, Inverse Probability Weighting will be modelled using the probability of having an EPC as a function of area-level data and demographics.

These analyses are exploratory and are intended both to quantify structural housing–health associations in the H3K Lothian cohort and to refine modelling approaches for the national study.

Intersectional multilevel analysis of individual heterogeneity and discriminatory accuracy (I-MAIHDA)

Health inequalities are the product of complex, multifaceted and interrelated exposures that are often experienced in combination as individuals occupy intersecting social hierarchies or environments.⁶⁴ Intersectionality analysis is a theoretically-oriented and driven approach that considers multiplicative interactions between intersectional strata to enable interrogation of the role of systems of oppression in the creation of inequity and subsequent inequalities.^{65,66}

Here, I-MAIHDA will be used to explore how social determinants of health intersect to shape ARI risk. Intersectional strata are; sex, ethnicity, SIMD quintile, tenure type and household income. A multilevel logistic regression model will be fitted, with the combination of inequality dimensions (intersectional strata) included as random effects, to test if intersectional strata contribute to between-stratum variance.⁶⁷ This will be determined by calculating the intraclass correlation coefficient (ICC). Partially-adjusted intersectional models, where intersectional strata are included as fixed effects, will then be fitted to determine which intersectional strata explain most of the between-stratum variance of the random effects. This will be determined by calculating the proportional change in between-stratum variance (PCV). Finally, the intersectional strata that defines the highest or lowest risk group will be calculated using pairwise comparisons between different groups.

Hidden Markov Model Development

A Hidden Markov Multistate Model (HMM) will be used as a complementary longitudinal approach to characterise respiratory health trajectories over early childhood. HMMs are particularly useful for studying childhood respiratory illness because infection burden is often reflected across multiple healthcare signals (e.g. hospital admissions, GP consultations and prescriptions), which individually provide only partial information about a child's underlying respiratory health status.

Markov processes consist of states, which represent underlying health status at a given time, and transitions, which represent movement between these states⁶⁸. An advantage of using the HMM is that rather than providing precise estimates, it uses a likelihood measure to construct a transition probability matrix, and an emissions matrix for probability of observations arising from a hidden state⁶⁸. Potential limitations to the application of the model are that it assumes that the probability of transition is based solely on the current health state and not previous health states, and the assumption that transitions between states are independent from time [for example as a result of a child growing up])⁶⁸.

In this study, respiratory health states are not directly observed, but are inferred from healthcare use patterns. The HMM therefore estimates the probability of different observed healthcare events arising from an underlying latent respiratory health state. These will include ARI-related hospital admissions, primary care consultations, and prescriptions or amoxicillin clarithromycin or salbutamol. The initial model will explore several transient respiratory health states representing increasing levels of respiratory illness burden (good, poor, poorer, and very poor respiratory health), and one absorbing

state; death from respiratory illness (Figure 4). Transitions between states will be modelled over time, allowing children to move between states as their respiratory health improves or worsens.

The model estimates transition intensities, representing the instantaneous probability of moving from one state to another. These transition probabilities are derived using maximum likelihood estimation and can be summarised in a transition intensity matrix. The HMMM also estimates an emissions matrix, describing the probability of each observed healthcare signal arising from each underlying latent health state. Energy efficiency (from EPC-derived SAP score) will be incorporated as a covariate in the transition model, alongside other child and environmental characteristics such as age, season, and neighbourhood context. This will allow exploration of whether children living in less energy-efficient homes spend more time in higher-burden respiratory health states or have an increased probability of transitioning into these states.

To avoid excessive model complexity, models with different numbers of latent states will be compared using standard goodness-of-fit criteria, including the Akaike Information Criterion (AIC). The model providing the best balance of statistical fit and interpretability will be selected for the main analysis. Model outputs will include estimates of time spent in each respiratory health state (sojourn time), transition probabilities between states, and the probability of different healthcare observations arising from each latent state.

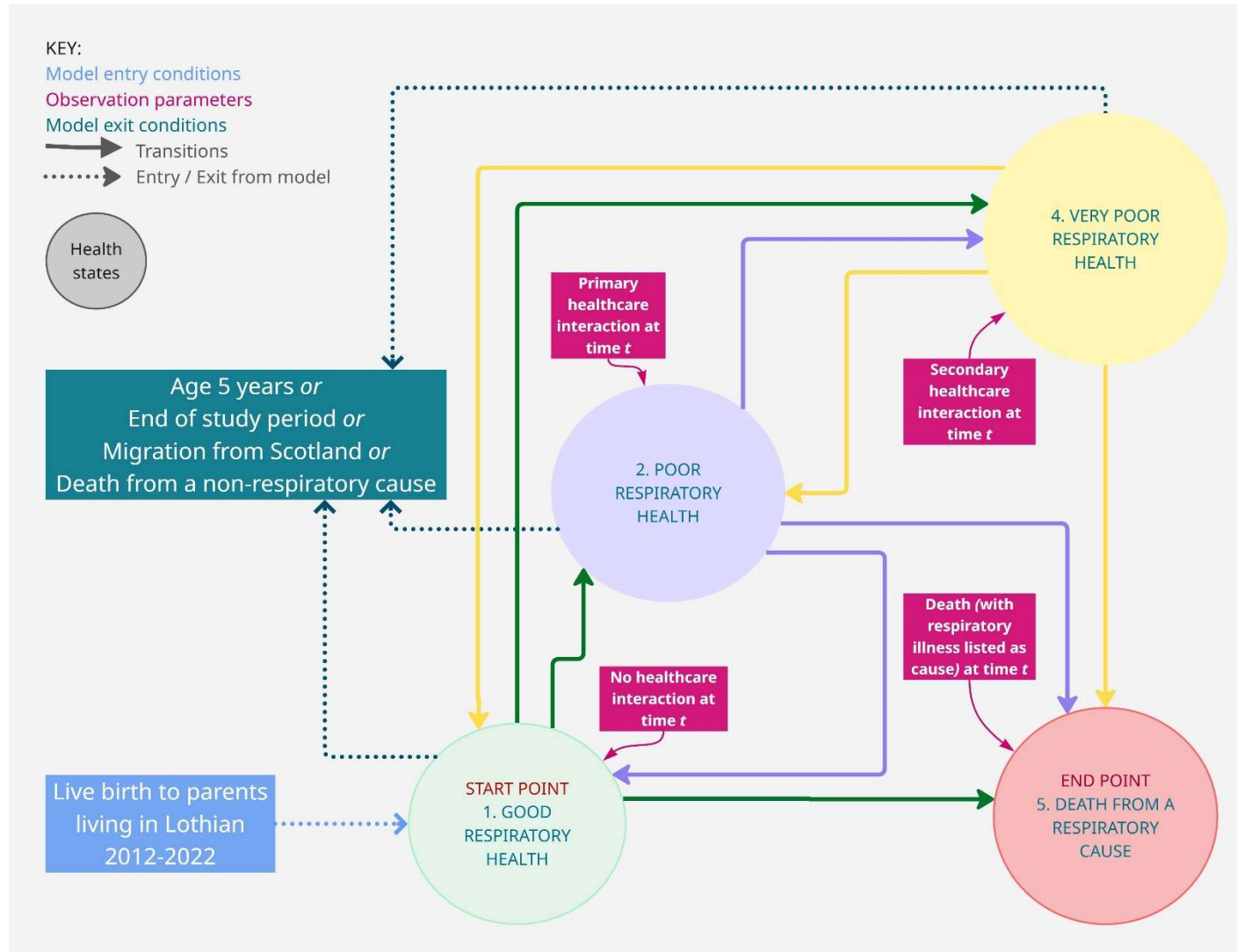


Figure 4: States, transitions and data sources for the Hidden Markov Model

Limitations of H3K Lothian dataset

The H3K Lothian cohort represents a regional dataset and therefore may not be fully representative of the wider Scottish population. Housing exposure measures are derived from the most recent EPC available for each property and do not capture historical changes in energy efficiency over time. This may introduce exposure misclassification where retrofits occurred prior to the recorded EPC inspection. In addition, the dataset does not include property-level energy consumption data or complete records of housing energy efficiency interventions, meaning that household energy deficit and underheating cannot be directly measured. Consequently, analyses undertaken in DataLoch focus on energy efficiency rather than underheating. These limitations are expected to result primarily in conservative estimates and will be addressed in the subsequent H3K Scotland cohort, which incorporates additional datasets including energy consumption and intervention records.

Patient and public involvement and engagement (PPIE)

Public involvement is central to the H3K programme. We work closely with a Parent Advisory Group made up of parents with lived experience of cold or poor-quality housing and its impact on children's health. The group has been involved since the inception of the study and contributes to the design, interpretation, and dissemination of the research. Their input has shaped project outputs including the H3K logo, a public-facing animation on cold homes, and an interactive educational game, which has also been tested with local primary school pupils to ensure accessibility and relevance.

Members of the Parent Advisory Group have contributed to study development, advised on key research questions and variables, and participated in public engagement activities including workshops and conference sessions on ethical storytelling and healthy homes. Our approach to collaboration is described as **"Beyond Remuneration"**, recognising lived experience as expertise in its own right. In addition to payment for their time, group members have the opportunity to gain formal qualifications through Media Education, including SCQF awards in media skills and community leadership.

Future PPIE activities will include continued collaboration with the Parent Advisory Group, expanding membership to include a wider range of perspectives, and sharing project outputs through public engagement events such as science festivals and community workshops. Project updates and resources are shared through the HHHK study website (<https://homesheatandhealthykids.github.io>).

Dissemination

Findings from the study will be disseminated to policy, academic, clinical, and public audiences. Results will be shared with policy stakeholders working in housing, energy, and public health through briefings, stakeholder meetings, and existing networks linked to the H3K programme. This will help ensure that emerging evidence on housing and child health can inform discussions around fuel poverty, housing quality, and energy efficiency policy.

Results will also be disseminated to the research and clinical community through peer-reviewed journal publications and presentations at national and international conferences. Short summaries of key findings will be shared through professional networks and online platforms to reach researchers, clinicians, and practitioners working across child health and housing. Project code and documentation will be made available on our public GitHub site <https://github.com/HomesHeatAndHealthyKids> to improve transparency and reproducibility.

For public and community audiences, accessible summaries of findings will be produced and shared through the project website and blog. These will also be circulated through local organisations

working on housing, fuel poverty, and child health in the Lothian region, helping to ensure that findings reach families, community groups, and practitioners who support households affected by cold homes. Where appropriate, outputs will also be shared through public engagement activities associated with the wider H3K programme.

Conclusion

The Homes, Heat and Healthy Kids study brings together linked health and housing data to better understand how children's home environments influence respiratory health in early life. By examining how structural housing conditions relate to respiratory infection burden in early childhood, this work will provide new evidence to inform housing, energy, and public health policy. Ultimately, the study aims to support efforts to ensure that homes are healthier environments for children and that policies addressing housing and fuel poverty consider their long-term impacts on child health.

Statement on conflicts of interest

OS is a member of the Scottish Fuel Poverty Advisory Panel, an independent body which advises the Scottish Government on Fuel Poverty Policy.

Funding

OS and TW are funded by a Wellcome Early Career Award [308485/Z/23/Z]. Hannah Law is funded by the Safe Deposits Scotland Charitable Trust. EH is funded by UKRI EPSRC [EP/Y030869/1]. CF is funded by ADR-UK.

Ethics statement

H3K Scotland was approved by the Edinburgh Medical School Research Ethics Committee [24-EMREC-081]. H3K Lothian was approved by the regional DataLoch Ethics Board [DL_2023_023].

Data availability statement

The data used in this study are accessed through secure Trusted Research Environments and contain sensitive, de-identified individual-level health and housing records. These data are not owned by the research team and cannot be shared directly. Access to the underlying data is subject to approval from the relevant data controllers and governance bodies, and researchers wishing to use the data must apply through the appropriate data access processes.

References

1. Al Rajeh AM, Naser AY, Siraj R, et al. Acute upper respiratory infections admissions in England and Wales. *Medicine* 2023; 102: e33616.
2. NHS Digital. Hospital Admitted Patient Care Activity 2021–22. National Statistics., <https://digital.nhs.uk/data-and-information/publications/statistical/hospital-admitted-patient-care-activity/2021-22> (2022).
3. Somerville M, Mackenzie I, Owen P, et al. Housing and health: does installing heating in their homes improve the health of children with asthma? *Public Health* 2000; 114: 434–439.
4. Simoes E, Cherian T, Chow J, et al. Acute Respiratory Infections in Children, <https://www.ncbi.nlm.nih.gov/books/NBK11786/> (2019).
5. Barnes M, Butt S, Tomaszewski W. The Dynamics of Bad Housing: The Impact of Bad Housing on the Living Standards of Children. National Centre for Social Research.
6. Fisk WJ, Singer BC, Chan WR. Association of residential energy efficiency retrofits with indoor environmental quality, comfort, and health: A review of empirical data. *Building and Environment* 2020; 180: 107067.
7. Antova T, Pattenden S, Brunekreef B, et al. Exposure to indoor mould and children's respiratory health in the PATY study. *Journal of Epidemiology & Community Health* 2008; 62: 708–714.
8. Kumar P, Mukherjee A, Randev S, et al. Effect of acute respiratory infections in infancy on pulmonary function test at 3 years of age: a prospective birth cohort study. *BMJ Open Respiratory Research* 2020; 7: e000436.
9. Gern JE, Rosenthal LA, Sorkness RL, et al. Effects of viral respiratory infections on lung development and childhood asthma. *Journal of Allergy and Clinical Immunology* 2005; 115: 668–674.
10. Mocelin HT, Fischer GB, Bush A. Adverse early-life environmental exposures and their repercussions on adult respiratory health. *Jornal de Pediatria*; 98. Epub ahead of print December 2021. DOI: 10.1016/j.jped.2021.11.005.
11. Allinson JP, Chaturvedi N, Wong A, et al. Early childhood lower respiratory tract infection and premature adult death from respiratory disease in Great Britain: a national birth cohort study. *The Lancet* 2023; 401: 1183–1193.
12. Justice NA, Le JK. Bronchiolitis, <https://www.ncbi.nlm.nih.gov/books/NBK441959/> (2023).
13. El-Azami-El-Idrissi M, Lakhdar-Idrissi M, Chaouki S, et al. Pediatric recurrent respiratory tract infections: when and how to explore the immune system? (About 53 cases). *Pan Afr Med J*; 24. Epub ahead of print 2016. DOI: 10.11604/pamj.2016.24.53.3481.
14. World Health Organization, United States, France, et al. (eds). *WHO housing and health guidelines*. Geneva, Switzerland: World Health Organization, 2018.

15. Stone J, Padley M. *A Minimum Income Standard for the United Kingdom in 2025*. Centre for Research in Social Policy, <https://www.jrf.org.uk/sites/default/files/pdfs/a-minimum-income-standard-for-the-united-kingdom-in-2025-1ba060e7fdde141dbdf6ef5e2a430606.pdf> (9 September 2025).
16. Fuel Poverty (Targets, Definition, and Strategy) (Scotland) Act 2019. *Fuel Poverty (Targets, Definition, and Strategy) (Scotland) Act 2019*, <https://www.legislation.gov.uk/asp/2019/10/contents>.
17. Scottish Government. *Scottish House Condition Survey: 2023 Key Findings*. Directorate for Tackling Child Poverty and Social Justice; Communities Analysis Division, <https://www.gov.scot/binaries/content/documents/govscot/publications/statistics/2025/01/scottish-house-condition-survey-2023-key-findings/documents/scottish-house-condition-survey-2023-key-findings/scottish-house-condition-survey-2023-key-findings/govscot%3Adocument/Scottish%2BHouse%2BCondition%2BSurvey%2B2023%2BKey%2BFindings.pdf> (January 2025).
18. Scottish Government. Heat in Buildings Strategy: Achieving Net Zero Emissions in Scotland's Buildings., <https://www.gov.scot/publications/heat-buildings-strategy-achieving-net-zero-emissions-scotlands-buildings/> (2021).
19. Niculita-Hirzel H. Latest Trends in Pollutant Accumulations at Threatening Levels in Energy-Efficient Residential Buildings with and without Mechanical Ventilation: A Review. 2022; 19: 3538.
20. Brambilla A, Sangiorgio A. Mould growth in energy efficient buildings: Causes, health implications and strategies to mitigate the risk. *Renewable and Sustainable Energy Reviews* 2020; 132: 110093.
21. Rodgers SE, Bailey R, Johnson R, et al. Emergency hospital admissions associated with a non-randomised housing intervention meeting national housing quality standards: a longitudinal data linkage study. *J Epidemiol Community Health* 2018; 72: 896–903.
22. Sharpe RA, Thornton CR, Nikolaou V, et al. Higher energy efficient homes are associated with increased risk of doctor diagnosed asthma in a UK subpopulation. *Environment International* 2015; 75: 234–244.
23. Sharpe RA, Machray KE, Fleming LE, et al. Household energy efficiency and health: Area-level analysis of hospital admissions in England. *Environ Int* 2019; 133: 105164.
24. Hamilton I, Milner J, Chalabi Z, et al. Health effects of home energy efficiency interventions in England: a modelling study. *BMJ Open* 2015; 5: e007298.
25. Tieskens KF, Milando CW, Underhill LJ, et al. The impact of energy retrofits on pediatric asthma exacerbation in a Boston multi-family housing complex: a systems science approach. *Environ Health* 2021; 20: 14.
26. Kulkarni R, Harrison E, Semple FF, et al. The Impact of Home Energy Efficiency Measures on Children's Respiratory Health: A Scoping Review. Epub ahead of print 2025. DOI: 10.17605/OSF.IO/XYEGB.

27. NICE. *Excess winter deaths and illness and the health risks associated with cold homes*. National Institute for Health and Care Excellence, <https://www.nice.org.uk/guidance/ng6/resources/excess-winter-deaths-and-illness-and-the-health-risks-associated-with-cold-homes-pdf-51043484869> (5 March 2015).
28. Scottish Government. *Scottish House Condition Survey: 2021 Key Findings*. Directorate for Tackling Child Poverty and Social Justice; Communities Analysis Division, <https://www.gov.scot/publications/scottish-house-condition-survey-2021-key-findings/documents/> (30 May 2023).
29. Tammes P, Sartini C, Preston I, et al. Use of primary care data to predict those most vulnerable to cold weather: a case-crossover analysis. *Br J Gen Pract* 2018; 68: e146–e156.
30. Clark D, Dibben C. Adding a Residential Dimension to the Scottish Population Spine – CHI-UPRN Residential Linkage (CURL). *International Journal of Population Data Science*; 7. Epub ahead of print 25 August 2022. DOI: 10.23889/ijpds.v7i3.1946.
31. Home | Dataloch, <https://dataloch.org/> (accessed 22 April 2026).
32. Public Health Scotland. SMR01 - General Acute Inpatient and Day Case. *SMR01*, <https://publichealthscotland.scot/resources-and-tools/health-intelligence-and-data-management/national-data-catalogue/national-datasets/search-the-datasets/general-acute-inpatient-and-day-case-scottish-morbidity-record-smr01/>.
33. Public Health Scotland. SMR02 - Maternity and Inpatient Day Case. *SMR02*, <https://publichealthscotland.scot/resources-and-tools/health-intelligence-and-data-management/national-data-catalogue/smr-data-manual/definitions-by-smr-record-section/smr02-maternity-inpatient-and-day-case/current-pregnancy-section/>.
34. Scottish Government. Energy Performance Certificates: guide, <https://www.gov.scot/publications/energy-performance-certificates-guide/>.
35. Department for Business, Energy & Industrial Strategy. The Government's Standard Assessment Procedure for Energy Rating of Dwellings (SAP 10.2), <https://www.gov.uk/guidance/standard-assessment-procedure> (2012).
36. Urban Tide. Summary Report. uZERO:Tackling the growing fuel poverty crisis with AI., <https://urbantide.com/uzero> (2022).
37. Data Communications Company. *Smart DCC*, <https://www.smartdcc.co.uk/about-dcc/> (accessed 22 April 2026).
38. National Archives. Output Area. *Census Geography*, <https://webarchive.nationalarchives.gov.uk/ukgwa/20220401215420/https://www.ons.gov.uk/methodology/geography/ukgeographies/censusgeography#output-area-oa>.
39. Scottish Government. Domestic Energy Performance Certificates, <https://statistics.gov.scot/data/domestic-energy-performance-certificates>.
40. Department for Energy Security and Net Zero. Subnational Electricity Consumption Data, <https://www.gov.uk/government/collections/sub-national-electricity-consumption-data>.

41. DWP. Stat-Xplore. *Stat-Xplore*, <https://stat-xplore.dwp.gov.uk/webapi/jsf/login.xhtml>.
42. Department for Energy Security and Net Zero. National Energy Efficiency Data-Framework (NEED) report: Summary of analysis using the National Energy Efficiency Data-Framework using consumption data from 2005 to 2021., <https://www.gov.uk/government/statistics/national-energy-efficiency-data-framework-need-report-summary-of-analysis-2023> (2023).
43. DEFRA. Modelled background pollution data, <https://uk-air.defra.gov.uk/data/pcm-data>.
44. WHO Global Air Quality Guidelines: Particulate Matter (PM2.5 and PM10), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide. 1st ed. Geneva: World Health Organization, 2021.
45. EPA. *Integrated Science Assessment for Particulate Matter*, <https://assessments.epa.gov/isa/document/&deid=347534> (2019).
46. EPA. *Integrated Assessment for Ozone and Related Photochemical Oxidants*, <https://www.epa.gov/isa/integrated-science-assessment-isa-ozone-and-related-photochemical-oxidants>.
47. EPA. *Integrated Science Assessment for Oxides of Nitrogen - Health Criteria*, <https://assessments.epa.gov/isa/document/&deid=310879> (2016).
48. Met Office, Hollis D, Carlisle E, et al. HadUK-Grid Gridded Climate Observations on a 1km grid over the UK, v1.3.1.ceda (1836-2024). 2025; 7269 Files, 379121117531 B.
49. Government S. Scottish Index of Multiple Deprivation 2020, <https://www.gov.scot/collections/scottish-index-of-multiple-deprivation-2020/> (2020).
50. Little R, Rubin D. *Statistical Analysis with Missing Data, Third Edition*. 1st edn. Wiley. Epub ahead of print 17 April 2019. DOI: 10.1002/9781119482260.
51. Scottish Government. A Health and Biomedical Informatics Research Strategy for Scotland, <https://www.gov.scot/publications/health-biomedical-informatics-research-strategy-scotland/pages/4/>.
52. Zhang H, Casey A, Guellil I, et al. FLAP: a framework for linking free-text addresses to the Ordnance Survey Unique Property Reference Number database. *Front Digit Health*; 5. Epub ahead of print 28 November 2023. DOI: 10.3389/fdgth.2023.1186208.
53. Ordnance Survey. Address Base Premium, <https://www.ordnancesurvey.co.uk/products/addressbase-premium>.
54. Raasveldt M, Mühleisen H. DuckDB: an Embeddable Analytical Database. In: *Proceedings of the 2019 International Conference on Management of Data*. Amsterdam Netherlands: ACM, pp. 1981–1984.
55. Wickham H. Tidy Data. *J Stat Soft*; 59. Epub ahead of print 2014. DOI: 10.18637/jss.v059.i10.
56. dbt labs. dbt documentation. *dbt documentation*, <https://docs.getdbt.com/>.

57. Kuan V, Denaxas S, Gonzalez-Izquierdo A, Direk K, Bhatti O, Husain S, Sutaria S, Hingorani M, Nitsch D, Parisinos C, Lumbers T, Mathur R, Sofat R, Casas JP, Wong I, Hemingway H, Hingorani A. Ear and Upper Respiratory Tract Infections. DOI: 10.48533/VEXZ-5631.
58. Rebecca M Joseph, Mohammad Movahedi, William G Dixon, Deborah PM Symmons. Respiratory Tract Infection. DOI: 10.48533/SJT3-H180.
59. Luke Daines, Ann Morgan, Mome Mukherjee, Mohammad Al Sallakh, Eimear O'Rourke, Jennifer K Quint. Asthma Primary care. DOI: 10.48533/0M80-3D37.
60. Jeremy I Hawker, Sue Smith, Gillian E Smith, Roger Morbey, Alan P Johnson, Douglas M Fleming, Laura Shallcross, Andrew C Hayward. Sore Throat. DOI: 10.48533/4NDJ-T563.
61. Chukwuma Iwundu, Clare MacRae, Eleftheria Vasileiou. Tonsillitis Primary care. DOI: 10.48533/TB1J-8S37.
62. Jeremy I Hawker, Sue Smith, Gillian E Smith, Roger Morbey, Alan P Johnson, Douglas M Fleming, Laura Shallcross, Andrew C Hayward. Otitis media. DOI: 10.48533/F585-AQ49.
63. Swann O, Williams TC, Fraser LK, et al. 586 Making paediatric research simpler across primary and secondary care datasets; harmonised codelists for chronic paediatric conditions across the UK. In: *British Association of General Paediatrics*. BMJ Publishing Group Ltd and Royal College of Paediatrics and Child Health, pp. A116–A117.
64. Merlo J. Multilevel analysis of individual heterogeneity and discriminatory accuracy (MAIHDA) within an intersectional framework. *Social Science & Medicine* 2018; 203: 74–80.
65. Evans CR, Leckie G, Subramanian SV, et al. A tutorial for conducting intersectional multilevel analysis of individual heterogeneity and discriminatory accuracy (MAIHDA). *SSM - Population Health* 2024; 26: 101664.
66. Leckie G, Bell A, Merlo J, et al. The Statistical Advantages of Multilevel Analysis of Individual Heterogeneity and Discriminatory Accuracy for Estimating Intersectional Inequalities. *Sociological Methods & Research* 2025; 00491241251385123.
67. Evans CR, Borrell LN, Bell A, et al. Clarifications on the intersectional MAIHDA approach: A conceptual guide and response to Wilkes and Karimi (2024). *Social Science & Medicine* 2024; 350: 116898.
68. Jackson CH. Multi-State Models for Panel Data: The **msm** Package for R. *J Stat Soft*; 38. Epub ahead of print 2011. DOI: 10.18637/jss.v038.i08.

